MMA3001 – Numerical Methods and Machine Learning

Individual Project Brief

Key points for students to start considering the project

25%

of the unit mark

Individual project

Source: MMA3001_2026_Project_Brief.pdf

OVERVIEW

What this project is assessing

Designed to assess the ability to apply computational techniques to meaningful engineering problems.

Numerical methods	**Machine learning**	**Computer vision**
Data analysis	**Optimisation**	**Combinations of techniques**

Assessment emphasis

- demonstrated individual learning
- quality and organisation of computational work
- validity of evaluation
- justification for technical decisions
- lessons drawn from results
- clear and professional communication

Project selection and open-ended scope

Selection route

- Select by the end of Week 6
- Choose a provided project and dataset
- Or propose an engineering problem and dataset
- Self-selected projects must be discussed before commencement

Approval considers

- suitability for the unit
- clearly defined inputs and outputs
- achievable scope
- suitable validation evidence
- data ownership, licensing and privacy
- credible alternative solutions for comparison

All projects are open-ended. The teaching team may ask for an overly complex proposal to be reduced or modified.

Possible project areas mentioned in the brief

Apple-tree reconstruction

- Input: RGB-D observations of apple trees collected over three seasons
- Output: estimated three-dimensional tree skeletons
- Validation: comparison with ground-truth skeletons

Frozen-meat packaging quality

- Input: RGB images of frozen-meat products and packaging
- Output: report describing meat quality and packaging quality
- Validation: comparison with labelled assessments or known inspection outcomes

Simulation-based projects

- Students may work on simulations rather than data analysis
- Discuss this when proposing your own project
- Details of sample topics will be provided separately

CORE REQUIREMENTS

Every project must include

Engineering problem

Clearly defined and relevant to mechanical or aerospace engineering

Technique

At least one numerical method and/or machine-learning technique

Evidence

Model development; validation and verification; sensitivity analysis or optimisation where appropriate; engineering interpretation

Workflow

Reproducible workflow using version control; GitHub required

Tools

Appropriate use of Python, Jupyter, numerical libraries and ML frameworks

AI use

Acknowledgement and details of AI usage

Documentation: sections to build around

1 **Engineering problem**

context, problem, why it matters, intended use, limitations of scope

2 **Inputs and outputs**

input format, units and domain; outputs, meaning and intended use; invalid/missing inputs

3 **Computational solution**

method, reason selected, assumptions, settings or parameters, implementation, limitations

4 **Alternative solutions**

at least one alternative or baseline solution; evidence rather than preference alone

5 **Validation**

what was validated, appropriate method, metrics or acceptance criteria, and limitations

6 **Optimisation and performance**

accuracy, runtime, memory, robustness, stability, settings, resolution and trade-offs

7 **AI use and reflection**

AI tools, contribution, checking, student decisions, errors, limitations, understanding and workflow

Computational approach and alternatives

Explain the selected method

- how the method works at an appropriate level
- why it was selected
- important assumptions
- settings or parameters
- how it was implemented
- known limitations

Compare alternatives or baselines

- accuracy, robustness and computational time
- memory use and implementation complexity
- interpretability and scalability
- suitability for the intended engineering application
- comparison is to show evidence-based selection

Validation is required

Every project must include a clear and justified validation method.

- known ground truth
- independent experimental data
- trusted benchmark
- cross-validation
- sensitivity and robustness testing
- withheld test data
- repeated measurements
- established computational method
- physical constraints or conservation relationships
- manual inspection with defined criteria

Explain: what was validated; why the method is appropriate; metrics or acceptance criteria; whether validation influenced model selection; and what validation cannot establish.

A numerical result or visually convincing output is not sufficient evidence of accuracy.

Optimisation and performance

Analyse and improve where appropriate

- accuracy
- runtime
- memory requirements
- robustness
- numerical stability
- model or algorithm settings
- input resolution
- data requirements
- repeatability
- scalability
- trade-offs

Use evidence to justify the final design

- make use of profilers and timing routines
- where appropriate, estimate arithmetic intensity and required FLOPS
- compare the original or baseline solution with the improved or selected solution
- justify the final design

AI use and reflection

Students may use AI tools throughout the project, but all material AI use must be disclosed.

- which AI tools were used
- what they were used for
- approximate level of AI contribution
- why AI was used for those tasks
- how AI-generated material was checked
- which important decisions remained the student's responsibility
- errors, limitations or unhelpful suggestions produced by AI
- how AI use affected understanding or workflow

Responsibility remains with the student

Submitted code, results, claims, references and engineering decisions must still be verified.

AI cannot be used to answer presentation and interview questions.

DELIVERABLE

GitHub repository: required contents

README file

orientation and workflow instructions

HTML source-code documentation

can be generated by pdoc or similar

Licensing agreement

included in the repository

Source code

functional main branch with inline comments and docstrings

Test functions and test report

can be generated by pytest

Accessible final link and zipped clone

link submitted; zipped clone submitted via Moodle

The repository must contain the code and documentation needed to review the computational work.

TIMELINE

Assessment deliverables and timing

1

End of Week 6

Select project after the mid-semester assessment

2

Remaining teaching weeks

Expected progress and project guidance discussed

3

Due dates

See Unit Overview Spreadsheet

4

Week 14 examination period

Individual presentation / demonstration

Written documentation: individual document of 5–10 pages maximum; references and appendices do not count towards the 10-page limit; submit via Moodle.

GitHub: accessible final repository link plus a zipped clone submitted via Moodle.

Presentation and interview

5 min

presentation

3 min

questions

Week 14 during the examination period

Presentation should include

- brief engineering problem: primary input and output
- GitHub and structure; code documentation
- code validation and testing evidence
- chosen computational approach and reason for choosing it
- principal limitations or lesson learned
- student use and evaluation of AI

Question themes may include

- represented physical or engineering system and assumptions
- influential parameters or boundary conditions
- numerical stability and convergence criteria
- sources of numerical error
- verification against analytical, manufactured or benchmark solutions
- vulnerability to numerical failure and mitigation

ASSESSMENT FOCUS

What strong work demonstrates

Problem & I/O

complete engineering context; inputs and outputs fully specified with format, units, domains, meaning and intended use

Repository

professional, clear, fully reproducible structure; README with complete orientation and workflow instructions

Code documentation

readable code; comments clarify intent; consistent docstrings; complete HTML documentation

Validation

appropriate validation design, correct metrics, comparisons and honest limitations

Approach justification

credible baseline, evidence-based method selection, quantitative analysis and trade-offs

AI reflection

complete disclosure, verification of AI content, identification of errors and clear evidence of understanding

A perfect solution is not required: a partially successful method can still demonstrate strong engineering work if it is validated honestly and its limitations, lessons and improvements are clearly explained.

Possible Open-Source Engineering Datasets

Dataset names and link labels are clickable. Students must still define a suitable engineering problem, inputs/outputs, method, baseline and validation plan.

Dataset	Possible engineering use	Source	Working link
Airfoil Self-Noise	Aeroacoustics / regression	UCI 291	Open official page
Combined Cycle Power Plant	Thermodynamics / power output	UCI 294	Open official page
Gas Turbine CO and NOx Emissions	Gas turbine emissions / energy yield	UCI 551	Open official page
Hydraulic System Condition Monitoring	Fluid power / component condition	UCI 447	Open official page
AI4I Predictive Maintenance	Manufacturing / machine failure	UCI 601	Open official page
Energy Efficiency	Building thermal load prediction	UCI 242	Open official page
Steel Plates Faults	Manufacturing quality classification	UCI 198	Open official page
MetroPT-3	Rail compressor predictive maintenance	UCI 791	Open official page
NASA C-MAPSS Jet Engine Simulated Data	Turbofan RUL / prognostics	NASA	Open official page

Note: NASA C-MAPSS is publicly accessible via NASA Open Data; UCI datasets above are listed on the UCI Machine Learning Repository.